

Reproducibility Guide, Receipts Map & Rubric Compliance

This document is written for judges and reviewers. It explains, in plain language, what is in the submission package, how to verify each headline number without any special access, and where every challenge requirement is satisfied. **Public code repository:** <https://github.com/sharadbachani-oss/merlin-quantum-vw-gic2026>

1 · What is in the package

FILE	WHAT IT IS
VW_REPORT_v2.md / Technical Report PDF	The 4-8 page technical report: method, motivation, setup, results, mandatory ablations, negatives
QUANTUM_ADVANTAGE_EXHIBIT.md	The standalone hardware-advantage exhibit (the 156-qubit adjudicated result)
RECEIPTS.md	The full claim → file → cloud-job-ID map (summarized in §3 below)
results/	The result JSONs behind every headline number, stored verbatim
fable_vw_rl.py	The quantum-in-the-loop RL training pipeline (arms A-E, one seed per invocation)
fable_a1p_flight.py	Exact-theorem anchor derivation and hardware-flight script (model gate runs on CPU)
fable_traffic_directed.py	Directed-mobility demonstration card (model gate runs on CPU)
verify.py / README.md / requirements.txt	Clean-environment replication: dependencies, seeds, hardware specs, verification commands

2 · Verify the headline numbers in minutes, no credentials needed

Every hardware figure regenerates from archived raw data. All grading and 'model gates' (the pre-flight exact checks) run credential-free on a laptop:

```
pip install numpy && python verify.py
python fable_a1p_flight.py model # exact-theorem anchors (statevector, CPU)
python fable_traffic_directed.py model # directed-mobility model gate (CPU)
python fable_vw_rl.py 21 A # RL baseline arm, seed 21 (CPU)
```

WHAT verify.py PRINTS	EXPECTED (REPORT VALUE)
Exact-theorem anchor waveform correlations (hardware vs theorem)	0.9836 (wall) / 0.9806 (null)
RL final quality at equal budget, held-out stress suite	baseline 67.1% vs curriculum 76.4% safe (+9.3 pts)
Framework flow-encoding, frozen prediction on device	-26% objective, 3.5x tighter spread
Jam-relaxation spectrum $S(q, \omega)$, real-time route	dominant collective line 0.0625 cyc/step ($q = 0.5-2.0$)
Safety certificate areas per seed (Dirac-trained vs Adam)	printed per seed; Adam-trained certificate ships (disclosed)

Re-running the quantum flights themselves (scout → fly → grade) requires the reviewer's own IBM Quantum and QCi Dirac-3 credentials; every flight script carries an open-plan spend guard and a frozen pre-registration file written before each hardware job.

3 · Claim → receipt map (summary; full map in RECEIPTS.md)

CLAIM	ARTIFACT	CLOUD RECEIPT
Advantage of route: $S(q, \omega)$ with no analytic-continuation step	results/vw_jam_relaxation_spectrum.json	IBM fez d9rm4j9dsedc73agrb70
Adjudicated classical boundary (two-sided hostile referee)	A1_FINAL_VERDICT (annex)	IBM kingston d9rdb1dsedc73agh5ng + attack receipts
RL: -35.4% rollout budget vs GRPO, 3/3 seeds	vw_rl_{A..E}_s{21,22,23}.json	71 QCi Dirac-3 job IDs (in JSONs)
RL: +9.3-pt robustness at equal budget; ~2x budget-equivalence; two-box	results/rl_finalquality_classical.json + box report	independent-machine replication

CLAIM	ARTIFACT	CLOUD RECEIPT
Safety: 78→100% under RTA; envelope predicted then confirmed; 22 μ s monitor	vw_rta_demo.json, vw_lyap_bias_sweep.json	3 seeds x 500 episodes x 2 machines
Flow encoding -26% / 3.5x (frozen framework prediction)	results/flow_encoding_test.json	12 paired Dirac-3 jobs
Exact-theorem anchors 0.98; derived 18-line spectrum beats detuned	results/a1p_result_20260808_200224.json	fez d9rm4j9dsedc73agrb70 (4/4 gates)
Negatives (compression; directed-traffic hardness; ensemble null)	\$7 of the report + result JSONs	same receipt class as every pass

4 · Rubric compliance matrix

CHALLENGE REQUIREMENT	WHERE IT IS SATISFIED
Functional quantum / QI component integrated in ≥ 1 VLAM stage	Quantum-in-the-loop curriculum selection inside GRPO training (§2 of report); TN-Lyapunov runtime-assurance monitor wrapping the learned driver (§3)
Quantitative results vs track baseline, mean $\pm$ std, ≥ 3 runs; $\geq 10\%$ efficiency gain	RL track: -35.4% rollout budget vs the named GRPO baseline (3 seeds, mean $\pm$ SD) and +9.3-pt robustness at equal budget - replicated on a second machine (bar: $\geq 10\%$)
4-8 page technical report with mandatory ablation isolating the QI component	Technical Report PDF: six-arm RL ablation (random / device / classical optimizer / tuned sampler / encoding variant) + safety rank ablation (r = 3 interior optimum)
Public code repository; README enables clean-environment replication	https://github.com/sharadbachani-oss/merlin-quantum-vw-gic2026 - verify.py replicates all results from archived data with numpy only; README lists dependencies, seeds, and hardware specs

Discipline note: every result carries a pre-registration frozen before execution, in-job controls, and ≥ 3 seeds; every negative result ships with the same receipt class as every pass - including a hostile classical attack commissioned against our own advantage claim, whose verdict we adopted verbatim. Team Merlin Digital · GIC 2026 dual-track finalist · <https://github.com/sharadbachani-oss/merlin-quantum-vw-gic2026>